

10.1 Introduction to Lidar

1. Product Description

Lidar is a remote sensing technology that uses laser beams to detect target characteristics such as position and velocity. It offers advantages including high measurement resolution, strong penetration ability, robust anti-interference capability, and effective stealth detection.

According to the measurement principle, Lidar can be classified into three types: triangulation Lidar, pulse Lidar, and coherent Lidar. The Lidar equipped on the MentorPi is of the pulse lidar type.



2. Lidar Working Principle

A Lidar system consists of a laser emission system, a scanning system, a laser receiving system, and a signal processing system.

First, the laser emission system sends out a detection signal (laser beam) toward the target. The scanning system scans the plane to gather planar information.

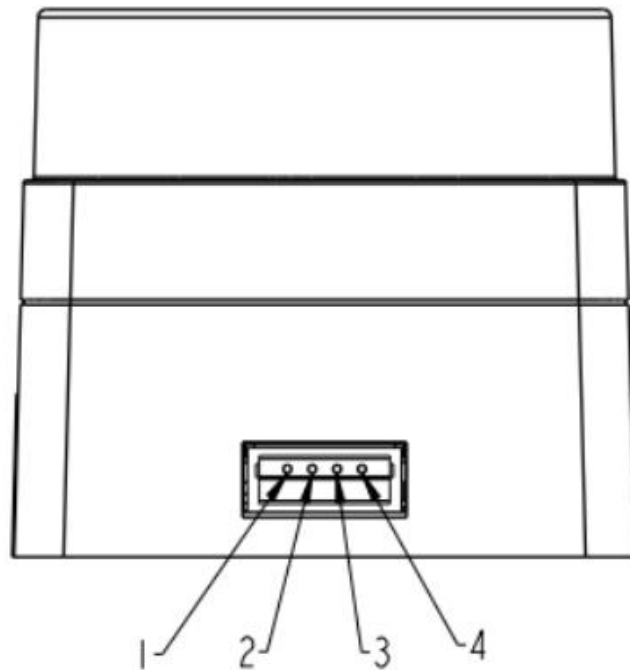
Next, the laser receiving system captures the laser beam reflected from the target object and generates a received signal.

Finally, the signal processing system processes the received signal to extract features such as the target's surface shape and physical properties (e.g., position, height, and velocity), thereby enabling the construction of a 3D model of the object.

3. Lidar Parameter

3.1 Communication Interface

The MS200 lidar connects to external systems via a connector for power supply and data transmission. For detailed interface definitions and parameter requirements, refer to the diagram/table below.



Pin	Signal	Attribute	Description
1	Tx	Serial data transmission	Tx (Transmitting from this device, 0V~3.3V)
2	Rx	Processing Incoming Data	Rx
3	GND	Power supply input (positive)	GND
4	VCC	Power supply input (positive)	DC 5V (4.5V~5.5V)

3.2 Electrical Parameter

Before use, it is essential to understand the operating conditions of the lidar.

Please ensure that the external power supply is properly designed or selected according to the Oradar MS200's input voltage range and peak startup current, to guarantee stable and reliable operation of the device.

Parameter	Minimum value	Typical value	Maximum value	Remarks
Operating power	4.5V	5.0V	5.5V	Pc power supply
Voltage ripple	-	-	100mV	
Startup current	-	-	400mA	
Operating current	180mA	230mA	280mA	
Operating temperature	-10℃	25℃	50℃	
Storage temperature	20℃	25℃	70℃	
Operating temperature	0	30%	80%	Non-condensing

3.3 Specification Parameter

Model	MS200:
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Type	Short-range
Distance Measurement Principle	TOF (Time of Flight) ranging, suitable for both indoor and outdoor applications
Scanning Angle	360°
Angular Resolution	0.8°
Measurement Frequency	4500 times per second
Scanning Frequency	7-15Hz
Dimensions	37.7*37.5*33mm
Weight	40g
Distance Measurement Accuracy:	3 m to 12 m: ± 20 mm
Light Source	905 nm laser
Measurement Range	White objects: 13.4 m Black objects: 13.4 m
Minimum	0.1m

Measurement Distance	
Ambient Light Resistance	30 KLux (illumination intensity)
Power Supply	DC5V±0.5V
Operating Temperature	Working: -10°C to 50°C Storage: -30°C to 70°C
Drive Method	Built-in brushless motor
Windows Support	Provides Windows-based PC software
ROS Support	ROS1/ROS2
Communication port	Standard asynchronous serial port (UART)